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Unraveling the interplay between mental workload, occupational fatigue, physiological responses and cognitive performance in office workers

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Recently, cognitive demands in workplaces have surged significantly. This study explored the intricate relationship among mental workload (MWL), occupational fatigue, physiological responses, and cognitive performance in office workers by using collective semi-parametric models. One hundred office workers were selected from twenty offices involved in cognitive performance. MWL was assessed through the NASA Task Load Index (NASA-TLX), and occupational fatigue was measured using the Persian version of the Swedish Occupational Fatigue Inventory. Physiological responses, including respiratory rate, the electrical conductivity of the skin (ECS), Heart Rate (HR), and other heart-related parameters, were recorded from the participants during a work shift. Selective and Divided Attention tests were chosen to evaluate workers' cognitive function based on cognitive task analysis. The mean of MWL and occupational fatigue scores were 66.28 ± 11.76 and 1.62 ± 1.07 , respectively. There was a significant moderate correlation between two dimensions, mental demand (0.429) and frustration (0.409), with functional fatigue. Also, Significant and, of course, nonlinear relationships were observed between MWL and HR ($R^2 = 0.44$, P-value < 0.001) and ECS ($R^2 = 0.45$, P-value < 0.001) and reaction time in selected ($R^2 = 0.34$, P-value < 0.001) and divided test ($R^2 = 0.48$, P-value < 0.001). Similarly, nonlinear relationships were observed between physiological responses and cognitive performance with fatigue among participants who had experienced higher levels of occupational fatigue. The MWL and fatigue seem to have a significant and non-linear effect on physiological parameters such as HR and ECS and cognitive parameters such as reaction time. Moreover, MWL can influence the dimension of functional fatigue of workers.

Keywords Fatigue, Neuropsychological tests, Cognitive performance, Office workers

Cognitive tasks have increased dramatically in recent years, and employees often experience a lot of cognitive demands in work settings¹. Cognitive tasks are the mental demands imposed on workers by attention, concentration, perception, memory, problem-solving, decision-making, and time limitations for performing parallel or series of multiple tasks. These mental processes are called cognitive performance. In work settings, cognitive performance in interaction with cognitive requirements should be managed. According to Kahneman's (1973) framework, managing performance necessitates a compensatory control system that dynamically allocates resources². Hence, this system allocates resources using an effort monitor and a supervisory controller capable

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of implementing various performance-cost trade-off modes. In this framework, performance can be maintained under pressure by accessing additional resources, but at the cost of higher subjective effort, and behavioral and physiological expenses. This can result in performance disruption in secondary tasks or the adoption of less effective methods, along with heightened psychophysiological arousal, stress, and fatigue consequences³. In other words, subjective demands can lead to cognitive workload. Cognitive load is the mental effort necessary for task execution, which can be impacted by task load and task design⁴. Although a reluctance for further effort is indicative of fatigue, however, In return, some studies have reported that mental workload (MWL) also negatively affects cognitive performance^{5,6}.

Prolonged engagement in cognitively challenging functions can result in mental fatigue, a multifaceted condition marked by a decreased willingness to exert effort and alterations in mood, drive, and cognitive processing⁷. Moreover, fatigue appears as a reduced physical⁸ or mental performance⁹ in a work. It is proven that high levels of job-induced fatigue can seriously affect workers' health¹⁰ and have negative consequences for workplace safety¹¹ and productivity¹². Fatigue (physical or mental) can impair cognitive function¹³ and process information¹⁴, slow down reaction time¹⁵, decline in attention or concentration¹⁶, limit short-term memory, disrupt judgment and decision-making¹⁷, and finally degrade motor performance¹⁸. As a result, fatigue can cause human errors and accidents¹⁹, ill health and injury²⁰, and reduced productivity²¹. Fatigue has been a root cause of major accidents¹¹. Therefore, fatigue needs to be managed, like any other harmful factors in the workplace²². Physical and mental consequences due to fatigue have a mutual relationship²³. Physical fatigue can impact mental processes⁸, mental tiredness can also affect physical abilities⁸. In work settings, fatigue can be associated with many factors, such as physically or cognitively demanding factors.

Moreover, maintaining cognitive performance over time requires additional resources and effort, which can lead to physiological costs. Research has indicated that physiological measures can often be considered indicators of MWL and subjective fatigue^{24–26}. Three systematic review studies have reported that the physiological parameters including heart rate variability (HRV), respiration rate (RR), electrodermal activity, and heart rate (HR) are the main indexes to measure MWL^{27–29}. HR and RR indicate immediate fluctuations in self-regulation and emotional states, representing the body's defense responses through physiological regulation. Electrodermal activity indicates the sympathetic nervous system activity by the eccrine glands. Therefore, these parameters are often considered as indicators of task engagement. As a consequence, in the present study, we used these measures.

Numerous studies have validated the association between mental workload, occupational fatigue, cognitive performance, and physiological response^{25,26,30}. Furthermore, some studies have found that measuring MWL and subjective fatigue during cognitive performance is another approach to investigating the link between mental fatigue and MWL⁸. Generally, a reciprocal association between mental workload and occupational fatigue has been difficult to separate because they can happen simultaneously, although they arise from different underlying processes. Some studies have not found a significant relationship between mental workload, occupational fatigue, physiological responses, and cognitive performance³¹. Remier and Mehler's³¹ found no correlation between physiological parameters and MWL. However, some of them have also explored the linear relationships between these variables. Studies conducted by Fallahi et al.²⁶, Hwang et al.³², Sosnowski et al.³³, and Shiet al³⁴ are examples of research that has identified a linear and direct relationship between physiological variables and cognitive workload. However, the study by Furlan et al.³⁵ and Lucini et al.,³⁶ revealed a linear yet indirect relationship between physiological parameters and MWL. Mehler et al. suggest that the MWL can unexpectedly increase the electrical conductivity of the skin (ECS), acting as an occupational stress factor³⁷. These ambiguities regarding the relationship between MWL and variables related to cognitive performance are also observed. Abdul Samad et al.³⁸ and Poulou et al.³⁹ observed that reaction time rises with a greater workload. Some believe an inverted U relationship exists between cognitive performance and mental workload^{40,41}. This theory highlights the importance of balancing mental workload and cognitive task demands to optimize cognitive functioning. As a result, the previous studies suggest a complex interplay among mental workload, occupational fatigue, physiological responses, and cognitive performance that warrants further investigation. Therefore, it is important to investigate whether there is a significant yet undetected non-linear relationship between physiological variables or cognitive performance parameters and mental workload. Using traditional linear regression techniques is the standard and easy approach to investigate the associations between physiological variables and mental workload/fatigue. However, the main restriction of this approach is the linearity assumption of the model, which is not true in many real cases. The utilization of non-linear statistical techniques in data analysis is crucial. These techniques aid in analyzing and modeling intricate and non-linear relationships within data, resulting in more precise and accurate data analysis. Furthermore, employing these techniques makes it feasible to enhance the data characteristics and model relationships that are challenging to analyze using linear statistical methods. Moreover, the application of non-linear statistical techniques in diverse fields like social sciences, biological sciences, medical sciences, and engineering facilitates the delivery of innovative and practical analyses. Although there are remedies like polynomial regressions or transformations of the variables, they can get complexities in the presence of clustered or longitudinally correlated data. A Multivariate Generalized Additive Mixed Model (GAMM), which accounts for the nonlinear relationships, has been proposed to address this issue.

Based on the current state of research studying this subject can elucidate the illustration of the relationship. As a result, the study was structured to examine the following three primary hypotheses in real work environments, through the application of collective semi-parametric models:

Hypothesis 1 The mental workload would have a significant but non-linear correlation with physiological response.

Hypothesis 2 The mental workload would have a significant but non-linear correlation with cognitive performance.

Hypothesis 3 The physiological responses and cognitive performance of fatigued individuals are different from those of non-fatigued individuals.

Materials and methods

Study population

This study involved 120 office workers selected from 20 office settings. One hundred and four workers agreed to participate by signing the informed consent, while 16 workers opted out of the study. In the following stage, the inclusion/exclusion criteria were checked. The inclusion criteria included not having neurological or cardiovascular diseases, sufficient sleep, good vision in both eyes, non-smoking, acceptable general health (assessed with the General Health Questionnaire (GHQ)), and work experience (WE) of over 1 year. Four participants did not meet the inclusion criteria. A total of 100 office workers took part in this study. They engaged in cognitive tasks such as complex financial calculations, data recording, and continuous customer interaction and worked approximately 8 h daily. The surveys and experimental parameters involving human subjects were carried out by the guidelines and regulations approved by the Institutional Ethics Committee.

Procedure

This study aimed to determine the intricate relationships between mental workload and fatigue as subjective judgments, cognitive function, and physiological variables as objective parameters. One hundred employees participated in this study. Every participant was examined during a daily work shift. Physiological variables and cognitive skills were measured in three periods: From 07:00 to 07:10, physiological parameters, and 07:10 to 07:15, mental skills. From 11:00 to 11:10, physiological parameters, and 11:10 to 11:15, cognitive skills. From 15:00 to 15:10, physiological parameters, and 15:10 to 15:15, mental skills. The participant sat quietly while the physiological measurements were performed. For this purpose, before beginning work, the biofeedback NeXus-4 device was attached to the participant's hand and chest, and the baseline physiological parameters were measured for 10 min. Then, the device was removed, and the participant was asked to conduct the cognitive tests for 5 min. Afterward, the participant was allowed to carry out daily tasks. This process was repeated at 11:00 and 15:00. At the end of the shift, the participant was asked to rate mental workload and work-induced subjective fatigue.

Mental workload measurement of participants

At the end of the work shift, the participants rated the perceived MWL using the NASA Task Load Index (NASA-TLX) software version 0.6 (Available at <http://www.nasatlx.com/>)⁴². It is a subjective and retrospective assessment of the mental burden they have encountered while working. MWL has six dimensions of cognitive, physical, and temporal requirements of the work, performance, effort, and frustration level⁴³. The rate of each measurement is within a 100-point range. There is a description for each dimension. Finally, the values of each measurement are weighted and computed to determine the score of MWL. A higher mental load score (closer to 100) indicates experiencing more cognitive workload.

Perceived fatigue measurement of participants

The occupational fatigue of participants was assessed using the Persian version of the Swedish Occupational Fatigue Inventory (P-SOFI)⁴⁴. All participants valued the occupational fatigue at the end of the work shift. The validity and reliability of this tool were confirmed at⁴⁴. The reliability coefficient (Cronbach's Alpha) and the internal consistency coefficient of the Persian version of the P-SOFI were 0.921 and 0.894, respectively⁴⁴. P-SOFI consists of 20 items related to fatigue such as lack of motivation, physical discomfort, and sleepiness⁴⁴. It is rated from not at all (zero) to very much (six). The total fatigue score is the sum of the three dimensions: functional (average of eight questions), physical (average of seven questions), and cognitive (average of five questions) dimensions. A higher total fatigue score (closer to 20) indicates more experienced fatigue.

Assessment of cognitive performance

First, the mental demands of office tasks were analyzed using cognitive task analysis (CTA). CTA is a method that can determine subtasks and cognitive demands for any occupation. Based on the CTA performed, the cognitive requirements were accuracy, attention, speed of action, visual monitoring, working memory, and detection of incorrect functions. Finally, the Selective and Divided Attention Test (SDAT) was selected to evaluate employees' cognitive function based on the CTA method. This test has been used to measure military, industry, and office workers' selective and divided attention⁴⁵. This test evaluates selective and divided attention individually. To assess selective attention, participants were seated at their workstations, where different visual stimuli were displayed on a laptop screen, prompting them to respond to specific stimuli. Likewise, to assess divided attention, participants were presented with two visual stimuli simultaneously on the screen, prompting them to respond to particular stimuli on either the right or left side of the screen.

The output of the SDAT test includes the number of false answers, no answers, correct answers, and reaction time. Reaction time is the time required to respond to the stimulus. The participants conducted the SDAT test at three time intervals: at the beginning (07:10 to 07:15), middle (11:10 to 11:15), and end (15:10 to 15:15) of the work shift.

Record of physiological parameters

Examined physiological parameters were the ECS, electrocardiogram (ECG), and respiration rate (RR). The participant's physiological parameters were recorded using a biofeedback device, Mind Media B.V. Nexus-4, at three time intervals: at the beginning (07:00 to 07:10), middle (11:00 to 11:10), and end (15:00 to 15:10) of the work shift. The device has Bluetooth capability for connecting to a laptop to receive and analyze data using the Bio Trace software (Ver. V2012C, Available at <https://www.mindmedia.com/en/products/biotrace-software/>).

The Nexus-4 records ECG using three AgCl electrodes: The negative electrode is attached to the right side of the chest, and the positive electrode to the left side of the sternum. The Bio Trace software was employed to extract HR and HRV (variation in beat-to-beat intervals in HR) from ECG signals, including parameters such as low frequency (LF: 0.04–0.15 Hz), high frequency (HF: 0.15–0.50 Hz), and LF/HF ratio⁴⁶. The Bio Trace software (Ver. V2012C) was also employed to remove ECG signal artifacts.

ECS was recorded using a skin conductance sensor attached to the participants' two fingers of the non-dominant hand. The respiration rate (RR) is measured via a belt around the chest. Finally, the percentage changes of the measured values to the baseline values were analyzed.

Data analysis

The GAMM was used to analyze the effect of MWL, Body Mass Index (BMI), age, WE, and time measurement on cognitive performance and physiological responses. The Generalized Estimating Equations Model (GEEM) was also used to analyze the relationship between cognitive performance and physiological responses to occupational fatigue. GAMM and GEEM are collective semi-parametric models. These statistical models consider both the flexibility of generalized additive models and random effects used in mixed effect models. The latter allows for considering non-linear relationships between predictors (BMI, etc.) and the outcome of interest (here, HR, etc.) by fitting smooth functions for each covariate. At the same time, the random effects are included in the model to capture the variability of correlated observations (multiple measurements for each subject). Considering Y_i as an outcome like HR and X_i as one of the covariates like age, a generalized additive mixed effects model can be written as:

$$Y_i = f(X_i) + \beta \times \text{Time} + Z_i b + \varepsilon_i$$

where the function f represents the relationships between each predictor variable and the response variable; and the term $Z_i b$ represents the random effects in the model (this accounts for the correlation between repeated measurements). In this model, the effect of time was considered linear (i.e. $\beta \times \text{Time}$ as we had three-time points for measuring the outcomes of the study). Also, ε_i represents the error term for the i th observation. In this study, to determine the functional form of predictor variables, the number of knots was tuned. The smoothness and roughness of the function depend on the knots as a degree of smoothness. We considered a range of knots (3–15) and selected the one for each variable that returned a better R^2 (the goodness of fit parameter) for the overall model.

Moreover, exploratory data analysis was conducted using clustering techniques (hierarchical method) for fatigue variables for further investigations. The silhouette value was utilized to determine the number of suggested clusters. The silhouette value measures how similar an object is to its cluster (cohesion) compared to other clusters (separation).

Ethics approval and consent to participate

This study was approved by the Ethical Committee for Research in Hamadan University of Medical Sciences (No. IR.UMSHA.REC.1400.639).

Results

The mean and standard deviation of the age, WE, and BMI of 100 participants from 20 offices were 39.13 ± 6.82 years, 14.87 ± 7.93 years, and 26.46 ± 3.74 kg/m², respectively. Moreover, the mean and standard deviation of scores of MWL and SOFI and their dimensions have been presented in Fig. 1.

The descriptive statistics of selected and divided attention tests and physiological parameters at the beginning, middle, and end of the work shift have been presented in Table 1. During the work shift, there are statistically significant variations in cognitive function and physiological parameters between the middle and the beginning or end.

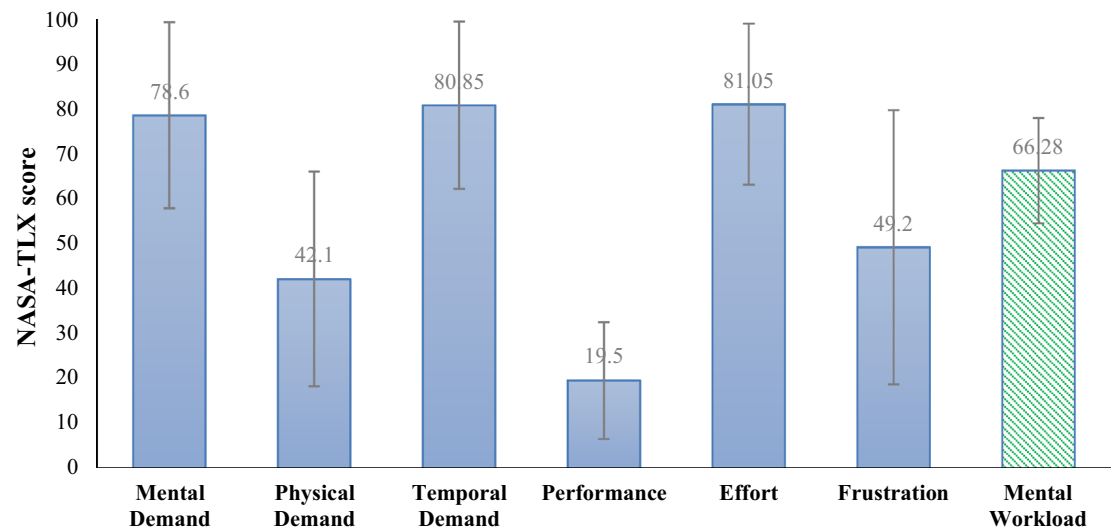
Table 2 shows the correlation between MWL and occupational fatigue (SOFI) and their dimensions.

Table 2 shows the results of investigating the statistically significant correlation between cognitive workload and occupational fatigue ($P < 0.05$). Also, according to the results, the correlation between cognitive workload and its several subscales (including mental demand, physical demand, temporal demand, and frustration) with occupational fatigue and its subscales was statistically significant ($P < 0.05$).

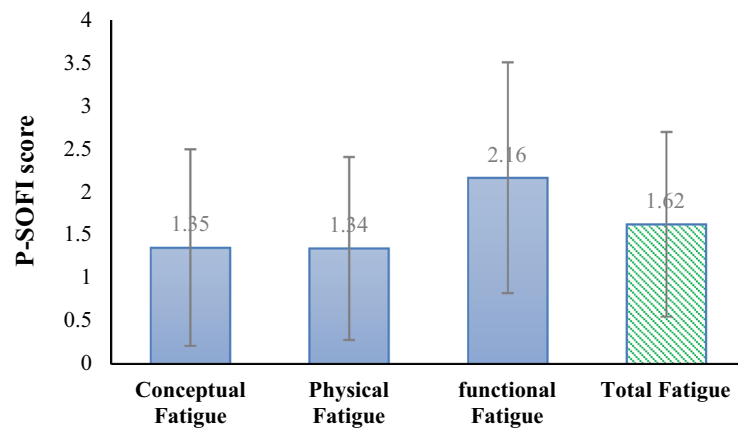
Table 3 shows the results of the GAMM in modeling the effect of variables of time (the time of measurement including beginning (07:00 to 07:15), middle (11:00 to 11:15), and the end (15:00 to 15:15) of the work shift), MWL, BMI, age, and WE on physiological parameters and cognitive performance.

This table includes regression coefficient, standard error, test statistics, and P-value for variables with linear effects and test statistics and P-value for nonlinear effects. In Table 3, the linear associations like the time variable were reported as regression coefficients. Instead, the functional regression coefficients for continuous variables are shown in Fig. 2 as they are not fixed values. Also, the determination coefficients were shown in Table 3 for various responses.

As seen for the HR, ECS, and reaction time (in both the selective and divided tests), the effect of time and MWL was statistically significant (controlling for MWL, age, BMI, and WE). Also, the association between some



a)



b)

Figure 1. The mean and standard deviation of (a) mental workload and six dimensions; (b) occupational fatigue and three dimensions in 100 office workers.

Parameters		Beginning of work shift	Middle of work shift	End of work shift
		Mean $\pm$ SD	Mean $\pm$ SD	Mean $\pm$ SD
Cognitive performance	Selected attention test			
	Number of false answers	4.50 $\pm$ 3.60	1.28 $\pm$ 1.52	3.24 $\pm$ 3.15
	Number of no answers	2.23 $\pm$ 2.47	1.61 $\pm$ 1.74	1.96 $\pm$ 2.63
	Percentage of correct answers	95.95 $\pm$ 3.03	96.32 $\pm$ 3.97	96.86 $\pm$ 2.95
	Reaction time (ms)	444.89 $\pm$ 44.04	432.30 $\pm$ 4.45	431.55 $\pm$ 46.85
	Divided attention test			
	Number of false answers	14.45 $\pm$ 8.67	3.30 $\pm$ 2.37	8.62 $\pm$ 5.88
	Number of no answers	6.95 $\pm$ 7.30	1.62 $\pm$ 1.74	3.66 $\pm$ 5.85
	Percentage of correct answers	85.13 $\pm$ 10.16	90.29 $\pm$ 7.92	91.78 $\pm$ 7.79
	Reaction time (ms)	629.80 $\pm$ 85.19	586.40 $\pm$ 79.92	569.51 $\pm$ 79.22
Physiological parameters	Heart rate (beats per minute)	80.21 $\pm$ 8.61	79.29 $\pm$ 9.63	78.71 $\pm$ 9.44
	Low frequency/high frequency	1.29 $\pm$ 0.29	2.89 $\pm$ 0.83	2.25 $\pm$ 1.01
	Respiration rate (breaths per minute)	22.98 $\pm$ 2.40	24.29 $\pm$ 2.42	23.54 $\pm$ 3.00
	Electrical conductivity of the skin (microsie-mens)	2.48 $\pm$ 1.91	3.78 $\pm$ 2.42	2.12 $\pm$ 2.15

Table 1. Descriptive statistics of cognitive performance (selected and divided attention tests) and physiological parameters at the beginning, middle, and end of the work shift.

Parameters	Analysis test	Functional fatigue	Physical Fatigue	Conceptual Fatigue	Total fatigue
Mental demand	Pearson correlation	0.429	0.392	0.376	0.390
	Sig. (2-tailed)	<0.001*	<0.001*	<0.001*	<0.001*
Physical demand	Pearson correlation	0.287	0.216	0.262	0.306
	Sig. (2-tailed)	0.004*	0.031*	0.008*	0.002*
Temporal demand	Pearson correlation	0.249	0.216	0.270	0.191
	Sig. (2-tailed)	0.013*	0.031*	0.007*	0.057
Performance	Pearson correlation	− 0.091	− 0.037	− 0.158	− 0.063
	Sig. (2-tailed)	0.370	0.718	0.117	0.531
Effort	Pearson correlation	0.089	− 0.021	0.192	0.095
	Sig. (2-tailed)	0.380	0.832	0.056	0.348
Frustration	Pearson correlation	0.409	0.396	0.353	0.350
	Sig. (2-tailed)	<0.001*	<0.001*	<0.001*	<0.001*
Mental workload	Pearson correlation	0.352	0.344	0.306	0.295
	Sig. (2-tailed)	<0.001*	<0.001*	0.002*	0.003*

Table 2. Correlation between mental workload (six dimensions) and occupational fatigue (three dimensions).

	Response variable	Parametric coefficients	Type of effect	Coefficient	Standard error	Test statistics	P-value	R ²
Physiological parameters	Heart rate	Time*	Linear	46.074	3.754	12.274	<0.001	0.44
		Mental workload	Nonlinear	−	−	3.280	<0.001	
		Age	Nonlinear	−	−	7.012	<0.001	
		Body Mass Index	Nonlinear	−	−	4.596	<0.001	
	Electrical conductivity of the skin	Time	Linear	1.54201	0.04816	32.02	<0.001	0.45
		Mental workload	Nonlinear	−	−	2.912	<0.001	
		Work experience	Nonlinear	−	−	3.422	<0.001	
		Body Mass Index	Nonlinear	−	−	4.522	<0.001	
Cognitive performance	Reaction time in divided test	Time	Linear	− 30.848	4.891	− 6.307	<0.001	0.48
		Mental workload	Nonlinear	−	−	4.898	<0.001	
		Body Mass Index	Nonlinear	−	−	4.129	<0.001	
	Reaction time in selective test	Time	Linear	− 10.306	3.375	− 3.054	0.002	0.34
		Mental workload	Nonlinear	−	−	3.240	<0.001	
		Work experience	Nonlinear	−	−	2.597	<0.001	
		Age	Nonlinear	−	−	7.766	<0.001	
		Respiration Rate	Nonlinear	−	−	5.448	0.020	
		Low frequency of HRV	Nonlinear	−	−	5.704	0.016	
		High frequency of HRV	Nonlinear	−	−	7.438	0.007	

Table 3. The results of the generalized additive mixed model in modeling neurophysiological parameters and cognitive performance variables. Time (the beginning (07:00 to 07:15), middle (11:00 to 11:15), and the end (15:00 to 15:15) of the work shift); *HRV* Heart Rate Variability. The negative regression coefficients indicate inverse/negative association while the positive values indicate direct/positive association.

parameters (Appendices A and B) and MWL was insignificant. The adjusted R^2 for HR was 0.44, indicating that about 44 percent of the variations observed in HR could be explained by the variables in the model.

The model's goodness of fit was checked visually as the “mgcv” package (Mixed GAM Computation Vehicle with Automatic Smoothness Estimation) has been provided (results were not shown). According to the plots, no serious violations from the normality of residuals were observed. The functional forms of variables with nonlinear associations are depicted in Fig. 2.

Based on the results, the fatigue level of office workers was classified into two clusters: low and high. These two groups were identified using the clustering method; so that the silhouette value of the two-cluster was greater compared to others (0.861). Low and high clusters' mean and standard deviation were 0.92 ± 0.43 and 2.81 ± 0.7 , respectively. As shown in Fig. 3, the mental and physical demands, frustration, and the score of MWL were significant in the two fatigue clusters. However, the data analysis did not show a significant difference between the two clusters' physiological data and fatigue-related cognitive parameters.

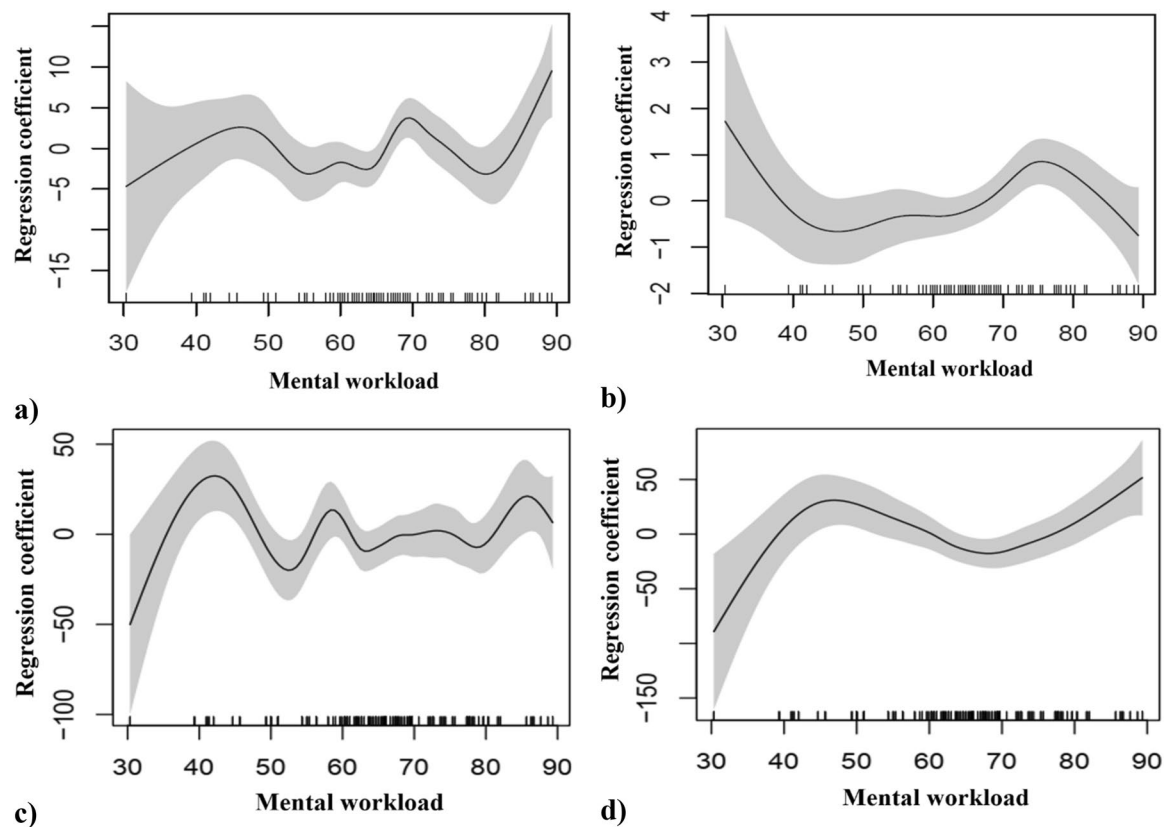


Figure 2. The functional form of the continuous variables in the fitted generalized additive mixed model predicts physiological (a) heart rate (HR); (b) electrical conductivity of the skin (ECS); and cognitive performance outcomes (c) reaction time in the selected attention test; (d) reaction time in the divided attention test.

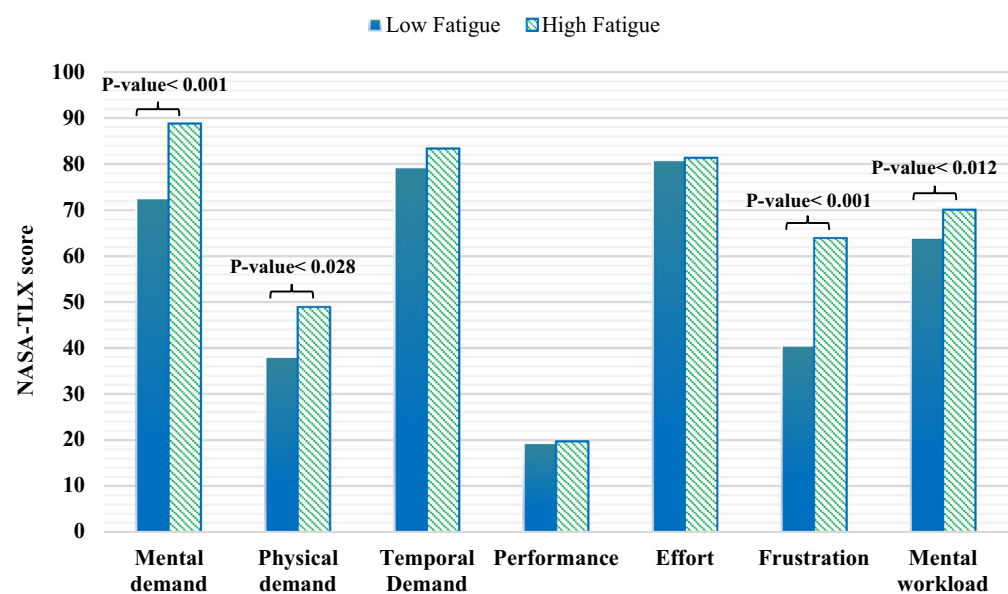


Figure 3. The mean of mental workloads and six dimensions in two groups of bank workers with low and high fatigue.

Therefore, the relationship between cognitive variables and physiological parameters with fatigue in two clusters was analyzed using the GEEM over two periods of time (T₁: From the beginning to the middle of the work shift; T₂: From the middle to the end of the work shift).

We considered including high fatigue as a dummy variable (1 = high fatigue, 0 = low fatigue) in the regression analysis. Table 4 presents the regression coefficients. Each coefficient indicates the average difference in physiological and cognitive variables between the high fatigue and low fatigue clusters over two time periods. For instance, the average skin electrical conductivity during the first period was 0.687 microsiemens higher in the high fatigue group compared to the low fatigue group.

Discussion

This study used collective semi-parametric models to investigate an intricate relationship among MWL, occupational fatigue, physiological parameters, and cognitive performance.

Mental workload and occupational fatigue of participants

The mean MWL of office workers was 66.28, which is classified as a high workload level according to the classification recommended category of Hart and Staveland⁴³. The primary components of MWL for participants included mental demands, temporal demands, and effort. In the study of bank staff, Darvishi et al. evaluated the MWL score as about 64.2⁴⁷. In another study, the minimum and maximum scores of the MWL of traffic control operators were reported to be approximately 62 and 79, respectively²⁶. Both studies determined mental demands and efforts as the main subscales of MWL. Thus, these studies have informed mental demands and efforts as the main subscales of MWL. As noted, the high MWL balances employees’ cognitive resources and task cognitive requirements. Therefore, there was not a balance between the participants’ cognitive resources and task requirements in this study. According to cognitive task analysis, participants’ requirements for accuracy, attention, action, visual monitoring, memory, and performance during their work shift could result in high mental and temporal demands, necessitating physical effort to sustain optimal performance.

The mean total score of occupational fatigue among participants was low (1.62 ± 1.07), with functional fatigue being the most effective subscale. In a study on occupational fatigue among nursing personnel, Ahmadi et al. found that they face high physical workload and fatigue⁴⁸. In another similar study, Vasilouee et al. confirmed that office workers experience high mental fatigue⁴⁹. Ahsberg et al. also noted that occupational fatigue is linked to cognitive concepts like sleepiness and lack of energy and motivation⁵⁰. Overall, the results confirmed that functional fatigue is influenced by the task’s nature and is connected to cognitive performance; decreased performance being the primary indicator of subjective fatigue among the participants.

Moreover, findings revealed that there was a statistically significant correlation between cognitive workload and its several subscales with occupational fatigue and its subscales. This finding agreed with the results of other studies. Vasilouee et al. have stated that there is a significant correlation between the Frustration dimension and workers’ occupational fatigue⁴⁹. Therefore, it can be concluded that workers’ MWL should be handled to diminish the employees’ fatigue.

Mental workload and physiological data

Hypothesis 1: Hypothesis 1 is supported. The mental workload had a significant but non-linear correlation with certain physiological responses (HR and ECS).

The findings revealed a significant and non-linear relationship between HR and ECS with MWL. However, no significant relationship with other physiological parameters was observed (Appendices A and B). A non-linear yet significant relationship was observed when considering the relationship between MWL and HR parameters.

Dependent variables		β [†]	SE*	P-value**
Physiologic parameters				
T ₁	Electrical conductivity of the skin (microsiemens)	0.687	0.230	< 0.001
	Heart rate (beats per minute)	3.052	1.543	0.040
T ₂	Low frequency/high frequency	-0.620	0.094	< 0.001
	Electrical conductivity of the skin (microsiemens)	-0.565	0.138	< 0.001
	Respiration rate (breaths per minute)	2.226	0.788	< 0.001
Cognitive performance variables				
T ₁	Number of correct answers in selected attention test	3.614	1.562	0.021
	Number of correct answers in selected attention test	0.682	0.337	0.043
T ₂	Number of no answers in selected attention test	-0.480	0.221	0.030
	Reaction time in divided attention test (s)	16.035	6.147	0.009

Table 4. The effect of high fatigue compared to low fatigue (regression coefficient) on physiologic parameters/ cognitive variables obtained by the Generalized Estimating Equations model. *SE stands for standard error; **P-values were adjusted for work experience, body mass index, marital status, education level, and mental workload; [†]the low fatigue category was considered as the reference. T₁: the changes from the beginning to middle of the work shift; T₂: the changes from the middle to the end of the work shift.

Several studies have confirmed the effect of MWL on heart-related parameters^{51,52}. Reimer et al. have found a clear connection between cognitive demands and cardiovascular responses³¹. In a meta-analysis study, Hughes et al. have found that heart rate variability is sensitive to the cognitive requirements of individual tasks⁵². However, contrary to current research findings indicating a significant nonlinear relationship between MWL and HR, some studies found no significant relationship. Remier and Mehler's assertion that there is no relationship between MWL and HR³¹. Kageyama et al. studied 223 healthy male white-collar workers to explore the connection between job stressors and daytime autonomic activities⁵³. They found no correlation between Heart parameters and five job stressor scores⁵³. A more detailed look at the study's findings reveals even more interesting points. The participants' HR increased significantly in the intervals of 0–45 units, 55–60, 65–70, and more than 80 of the MWL. This part of the findings was in line with the studies conducted by Fallahi et al.²⁶ Hwang et al.³², Sosnowski et al.³³, and Shi et al.³⁴; meaning that a higher workload lead to a higher physiological workload. Psychological stress can impact the body through two main pathways: the hypothalamic–pituitary–adrenal axis and the parasympathetic (PNS)/sympathetic nervous system (SNS)⁵⁴. According to the general stress theory—General Adaptation Syndrome (GAS)—described by Selye⁵⁵, the cognitive workload can be considered as stress in which the human body initiates changes in the endocrine system, physiological parameters, etc.²⁷. Hence, the body's biological response to stress may manifest as an elevated HR through the sympathetic system. The results indicate that in intermittent intervals, as workload increases, individuals' HR has increased; however, this does not necessarily mean that an absolute upward trend can be found in these intervals. In other words, an individual with an MWL of about 65 does not necessarily have an HR rate compared to an individual with an MWL of 45. Individuals experienced lower HR during certain intermittent parts despite an increased MWL. In other words, the participants' HR showed a significant decrease in the intervals of 45–55, 60–65, and 70–80 of the MWL. Along with these findings, Furlan et al.³⁵, and Lucini et al.³⁶ believe that individuals under chronic occupational/psychosocial stress will experience a decrease in HR. The findings suggest that a direct and linear correlation between mental workload and cardiac activity (HR in this context) is not evident. The connections between these two factors are intricate and influenced by multiple variables. Psychological stressors, such as mental workloads, lead to diverse individual reactions in the autonomic system. Berntson stated that individuals experiencing psychological stressors may display sympathetic activation, vagal withdrawal, or a reciprocal autonomic response pattern⁵⁶. These findings highlight the varied responses of individuals to psychological stressors. Krantz et al. believe that individual differences can also cause differences in the perception of stress and workloads and physiological changes⁵⁷. Therefore, individual differences in coping with stressful conditions, such as mental workload, can be considered as a significant factor in job and task allocation.

This study's relationship between the MWL and the ECS was nonlinear and significant. The findings also showed that as the MWL score increased from 45 to 75, ECS rose gradually. As stated, the ECS is an indicator of the activity of eccrine sweat glands that is controlled by the sympathetic nervous system. ECS is an indicator of stress; as the stress level increases, it increases noticeably⁵⁸. Various studies have concluded that MWL can significantly increase the ECS level. Mehler et al. believe that MWL is an occupational stress factor that can increase the ECS surprisingly³⁷. Furthermore, Berguer et al. have stated that as the complexity of tasks increases, the ECS level also increases significantly⁵⁹. The ECS is the only physiological index dependent on sympathetic activity, not parasympathetic⁶⁰. Thus, a decrease in ECS in high workload demonstrates a decrease in the function of the sympathetic nervous system⁵⁸. However, there was a declining trend in the ranges below 45 and above 75. This indicates that individuals with MWL below 45 may not necessarily have ECS. The results of Charles and Nixon's review study showed that none of the physiological characteristics alone are sufficient to differentiate the level of cognitive workload²⁷. Like cardiac parameters, ECS is also influenced by various factors simultaneously, including individual stress levels, individual differences, and stress adaption levels, which can well justify the non-linear relationship with mental workload.

In summary, the relationship between cognitive workload and physiological parameters (such as HR and ECS) is not linear; meaning that the body does not necessarily perceive an increase in workload as a stress that would result in proportional physiological changes with the increase in cognitive workload. Furthermore, cognitive workload has a non-linear and variable relationship with physiological parameters. This implies that each level of workload creates different stress levels in different individuals and triggers different physiological responses in individuals.

Mental workload and cognitive performance

Hypothesis 2: Hypothesis 2 is supported. The mental workload had significant but non-linear correlation with certain cognitive performance (reaction time).

As noted, based on the cognitive analysis, employees' cognitive function was evaluated by a divided and selective attention test. The results revealed a significant correlation between reaction time and MWL; other cognitive performance variables were insignificant. The reaction time changes in selected and divided attention tests fluctuated and could not be observed in a linear trend. This means that with an increased mental workload, it is not necessarily possible to observe a decrease in reaction time or performance. Furthermore, no U-shaped relationship similar to what many researchers have elucidated⁶¹ can be observed. Abdul Samad et al.³⁸ and Pouliou et al.³⁹ have also reported increasing reaction time with the increasing workload in the study on aircraft maintenance personnel and drivers. However, some researchers believe that a medium MWL is the best range to increase the reaction time of employees^{40,41}. Moreover, Northoff and Tumati have stated that the average is good, and the extremes are bad⁶². This suggests that as MWL increases, cognitive performance initially improves, but eventually, further increases in workload lead to a decline in performance. The connection between MWL and mental performance (reaction time of selected and divided attention test) is not linear, indicating that increased workload does not always result in decreased cognitive performance.

Occupational fatigue, physiological parameters, and cognitive performance variables

Hypothesis 3 The physiological responses and cognitive performance of fatigued individuals are different from those of non-fatigued individuals.

Preliminary data analysis showed that employees' occupational fatigue scores can be classified into two optimal clusters with significant differences, including low and high fatigue. Therefore, the results demonstrated that the average changes of ECS and HR in T_1 (from the beginning to middle of the work shift) for participants who experienced high fatigue were significantly higher than those who reported low fatigue (Table 4). These results in T_1 were completely aligned with the study of Olson et al.⁶³, Williamson et al.⁶⁴, and Mohanavelou et al.⁶⁵. Furthermore, Zhang and Yu and Jeklin have reported a strong relationship between a cardiac parameter (HRV) and fatigue^{25,66}. Also, the skin's resistance decreases (and conductivity increase) in a person who is fatigued, this occurs because of the rise in sweat gland secretion, which in turn enhances the electric current flow⁶⁵. Therefore, physiological parameters in the first half of the shift favored the expansion of fatigue; so that among individuals who had experienced more occupational fatigue, these parameters were significantly higher than those of non-fatigued individuals. In the second half of shift (T_2), the average LF/HF and ECS changes of participants with high fatigue were lower than those with low fatigue, versus the average changes of RR in participants with high fatigue were more than those with low fatigue (Table 4). Patel et al. found that LF/HF decreases with increasing fatigue⁶⁷. Mohanavelu et al. stated that when a person is fatigued, their breathing rate is higher than normal⁶⁵. Explaining the results related to ECS in the second half of the shift, which contradict the changes in LF/HF and RR, is quite challenging; However, numerous studies have recommended that the results of the ECS may be influenced by factors, such as temperature, humidity, time, age, and metabolism rate^{68,69}. However, generally, a significant difference in the physiological parameters under investigation is observed between fatigued individuals and non-fatigued individuals.

In first and second half of shift (T_1 and T_2), the mean "number of correct answers" in the selective attention test was significantly higher among employees with high fatigue! Generally, fatigue can greatly impact cognitive functions like verbal fluency, creativity, and planning skills⁷⁰. It is especially harmful for tasks that require focused attention rather than automatic responses. Unlike several previous studies, the varying outcome of the current study could be attributed to the internal motivation or elevated stress levels (linked to their arousal level) of individuals with greater fatigue experience.

However, in second half of shift (T_2), the number of "no answers" in the selective attention test was lower in the more fatigued group. Additionally, in T_2 the "reaction time" in the divided attention test was higher among participants with high fatigue. Various studies have demonstrated a link between fatigue and either increased reaction time²⁴ or diminished attention resources⁷¹. Borragán et al. asserted that fatigue is induced by the workload related to the available processing time⁷¹. In line with Borragán et al. finding, the current results support the theory of a restricted attentional resource system and its connection to fatigue. Overall, the study results do not suggest that individuals experiencing higher levels of fatigue significantly display poorer cognitive performance variables compared to those who are not fatigued; Furthermore, certain factors may impact the association between fatigue and cognitive performance, which should be carefully examined in future research.

During this study, there were some limitations. We attempted to elucidate the nature of complex relationships between the outcomes of MWL and physiological parameters, cognitive performance and occupational fatigue. However, due to their mutual influence on each other, further research is needed in this area to provide a clearer understanding of the relationships between them. Moreover, the physiological and cognitive data were measured in discrete times, therefore we could evaluate only linear effects for time. Additionally, since this study was conducted in a real-world work environment, there may be intervening factors, such as psychosocial and environmental conditions, that could have influenced the current study's findings. Furthermore, all participants were male, and the females were not studied. Another obstacle was that this study was carried out cross-sectional; there is no precise basis for establishing causality in cross-sectional studies. Thus, this problem should be addressed in future studies.

Conclusion

The study attained important insights into unraveling the interplay of MWL, occupational fatigue, physiological response, and cognitive performance in office workers. Mental and temporal demands and effort were the primary factors contributing to a high MWL among office workers. While the study showed that workers had a low level of occupational fatigue, "functional fatigue" emerged as the main component of occupational fatigue. There is a notable correlation between mental demands (and its various aspects) and occupational fatigue, including its different components.

Cognitive workload has generally been high among employees and has shown a significant but non-linear relationship between physiological parameters and cognitive performance. Moreover, the physiological responses and cognitive performance of fatigued individuals are different from those of non-fatigued individuals. Therefore, employers in work environments with high cognitive demands must take control measures to reduce MWL and fatigue to optimally maintain employees' physical health and cognitive performance. It is also recommended that considering the non-linear relationship between MWL and physiological parameters/cognitive performance variables, which is more pronounced in fatigued individuals, particular attention should be paid to individual cognitive differences in assigning jobs with high cognitive demands.

Data availability

The data that support the findings of this study are available on request from the corresponding author upon reasonable request.

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Author contributions

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Competing interests

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Additional information

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